Computational Mechanics

3. Representing the past

Solutions

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Instructions

These notes reproduce each exercise before its solution. Equivalent arguments, diagrams, and notation should receive full credit when the mathematical reasoning is correct.

Exercise 1

Translating between HMM diagrams and matrices

For an edge-emitting hidden Markov model, use the convention

$$T_{ij}^{(a)} := \Pr(X_{t+1} = a, S_{t+1} = j \mid S_t = i).$$

Thus rows index the current hidden state, columns index the next hidden state, and the superscript records the emitted symbol. Throughout this exercise, $0 < p, q < 1$.

(a) Suppose that $T_{ij}^{(a)} = r$. Explain each component of the following diagram:

$$i \xrightarrow{a:r} j.$$

What does a zero entry mean? Why is the matrix $T := \sum_{a \in \mathcal{X}} T^{(a)}$ row-stochastic?

(b) **From matrices to a diagram: Zero–One–Random.** Take the state order to be $(0, 1, R)$ and suppose that

$$\eta^{(\emptyset)} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}, \qquad T^{(0)} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1-p & 0 & 0 \end{bmatrix}, \qquad T^{(1)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ p & 0 & 0 \end{bmatrix}.$$

(i) Draw the corresponding edge-emitting HMM. Describe informally the three phases represented by 0, 1, and R.

(ii) Calculate $\Pr(X_{1:3} = 011)$ and $\Pr(X_{1:3} = 000)$ in two ways: first by tracing paths through your diagram, and then from

$$\Pr(X_{1:3} = x_{1:3}) = \eta^{(\emptyset)} T^{(x_1)} T^{(x_2)} T^{(x_3)} \mathbf{1}.$$

(c) **From a diagram to matrices: the Even Process.** Consider the edge-emitting HMM

$$A \xrightarrow{0:q} A, \qquad A \xrightarrow{1:1-q} B, \qquad B \xrightarrow{1:1} A,$$

with no other edges. Take the state order to be (A, B) and let $\eta^{(\emptyset)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$.

(i) Construct $T^{(0)}$ and $T^{(1)}$. Account explicitly for every zero entry in the two matrices.

(ii) Form $T = T^{(0)} + T^{(1)}$ and verify that it is row-stochastic.

(iii) Calculate $\Pr(X_{1:3} = 011)$ and $\Pr(X_{1:3} = 010)$ both by tracing paths through the diagram and by multiplying the appropriate symbol matrices. Explain how the second calculation reflects the defining constraint of the Even Process.

(d) **Extension.** Let $\mathcal{X}$ be the observation alphabet of an arbitrary finite-state HMM, let $T = \sum_{a\in\mathcal{X}} T^{(a)}$, and let $\eta^{(\emptyset)}$ be a probability distribution. Prove that, for every $n \geq 0$,

$$\sum_{w\in\mathcal{X}^n} \Pr(X_{1:n} = w) = \eta^{(\emptyset)} T^n \mathbf{1} = 1.$$

Solution.

(a) In the diagram, i is the current hidden state, j is the next hidden state, a is the emitted symbol, and r is the probability of that joint emission-transition event conditional on starting in i. Thus the edge records $T^{(a)}_{ij} = \Pr(X_{t+1} = a, S_{t+1} = j \mid S_t = i) = r$. A zero entry means that no such emission-transition pair is possible. For a fixed current state i, the events indexed by all pairs (a, j) exhaust the possible emitted symbols and next states. Consequently,

$$\sum_{a\in\mathcal{X}} \sum_j T^{(a)}_{ij} = 1.$$

Since

$$T_{ij} = \sum_{a\in\mathcal{X}} T^{(a)}_{ij},$$

every entry of T is nonnegative and its ith row sums to

$$\sum_j T_{ij} = \sum_j \sum_{a\in\mathcal{X}} T^{(a)}_{ij} = 1.$$

Hence T is row-stochastic.

(b) (i) The nonzero entries give the transitions

$$0 \xrightarrow{0:1} 1, \qquad 1 \xrightarrow{1:1} R, \qquad R \xrightarrow{0:1-p} 0, \qquad R \xrightarrow{1:p} 0.$$

Thus the diagram is a three-state cycle, with two differently labelled edges from R to 0. State 0 is the phase that emits 0 deterministically, state 1 is the phase that emits 1 deterministically, and state R is the random phase, which emits 1 with probability p and 0 with probability $1-p$.

(ii) For 011, the only possible initial state is 0. The model then follows

$$0 \xrightarrow{0:1} 1 \xrightarrow{1:1} R \xrightarrow{1:p} 0.$$

Including the initial probability $1/3$ gives $\Pr(011) = p/3$. Matrix multiplication gives the same result:

$$\eta^{(\emptyset)} T^{(0)} T^{(1)} T^{(1)} = \begin{bmatrix} p/3 & 0 & 0 \end{bmatrix}, \qquad \Pr(011) = p/3.$$

No state permits a path labelled 000: starting from 0, the first 0 moves to state 1, which cannot emit 0; starting from 1, even the first 0 is impossible; and starting from R, the first two zeros move through $R \to 0 \to 1$, after which a third zero is impossible. Correspondingly,

$$\eta^{(\emptyset)} T^{(0)} T^{(0)} T^{(0)} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}, \qquad \Pr(000) = 0.$$

(c) (i) In state order (A, B), the matrices are

$$T^{(0)} = \begin{bmatrix} q & 0 \\ 0 & 0 \end{bmatrix}, \qquad T^{(1)} = \begin{bmatrix} 0 & 1-q \\ 1 & 0 \end{bmatrix}.$$

In $T^{(0)}$, the entry (A, B) is zero because no 0-edge connects A to B, while the entire B row is zero because state B cannot emit 0. In $T^{(1)}$, the diagonal entries are zero because neither 1-edge is a self-loop. The two off-diagonal entries record $A \xrightarrow{1:1-q} B$ and $B \xrightarrow{1:1} A$.

(ii) We have

$$T = \begin{bmatrix} q & 1-q \\ 1 & 0 \end{bmatrix}.$$

Its entries are nonnegative and both rows sum to one.

(iii) The word 011 can be produced only by starting from A and following

$$A \xrightarrow{0:q} A \xrightarrow{1:1-q} B \xrightarrow{1:1} A.$$

Thus

$$\Pr(011) = \frac{1}{2} q(1-q).$$

The matrix calculation is

$$\eta^{(\emptyset)} T^{(0)} T^{(1)} T^{(1)} = \begin{bmatrix} \frac{1}{2} q(1-q) & 0 \end{bmatrix}, \qquad \Pr(011) = \frac{1}{2} q(1-q).$$

The word 010 would require the path to emit a 0 immediately after the transition $A \xrightarrow{1} B$, but state B cannot emit 0. Hence there is no such path. Equivalently,

$$\eta^{(\emptyset)} T^{(0)} T^{(1)} T^{(0)} = \begin{bmatrix} 0 & 0 \end{bmatrix}, \qquad \Pr(010) = 0.$$

This is the shortest example of an odd run of 1s occurring between two 0s, which the Even Process forbids.

(d) Expanding the product of the sum of the symbol matrices gives

$$T^n = \left(\sum_{a \in \mathcal{X}} T^{(a)} \right)^n = \sum_{w \in \mathcal{X}^n} T^{(w_1)} \cdots T^{(w_n)}.$$

Multiplying on the left by $\eta^{(\emptyset)}$ and on the right by $\mathbf{1}$ therefore yields

$$\eta^{(\emptyset)} T^n \mathbf{1} = \sum_{w \in \mathcal{X}^n} \eta^{(\emptyset)} T^{(w_1)} \cdots T^{(w_n)} \mathbf{1} = \sum_{w \in \mathcal{X}^n} \Pr(w).$$

Since T is row-stochastic, $T\mathbf{1} = \mathbf{1}$ and hence $T^n \mathbf{1} = \mathbf{1}$. Finally, $\eta^{(\emptyset)} \mathbf{1} = 1$, because the initial vector is a probability distribution. The required sum is therefore one.

Exercise 2

Belief states and a lossy next-token readout

Continue with the Zero–One–Random HMM from Exercise 1, using state order $(0, 1, R)$. For an allowable history $x_{1:n} = x_1 \cdots x_n$, compute its belief from scratch as

$$\eta^{(x_{1:n})} := \frac{\eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)}}{\eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)} \mathbf{1}}.$$

Here $\eta_i^{(x_{1:n})} = \Pr(S_n = i \mid X_{1:n} = x_{1:n})$. For an allowable extension $x_{1:n+1} = x_{1:n}a$, update recursively as

$$\eta^{(x_{1:n+1})} = \frac{\eta^{(x_{1:n})}T^{(a)}}{\eta^{(x_{1:n})}T^{(a)}\mathbf{1}}.$$

(a) Compute $\eta^{(0)}$ and $\eta^{(1)}$. Interpret their components as posterior probabilities over the three hidden phases.

(b) Show that the seven beliefs

$$\mathcal{B} = \{\eta^{(\emptyset)}, \eta^{(0)}, \eta^{(1)}, \eta^{(00)}, \eta^{(01)}, \eta^{(10)}, \eta^{(11)}\}$$

are distinct and that $\mathcal{B}$ is closed under every allowable symbol update. Draw the symbol-labelled transition diagram on $\mathcal{B}$.

(c) Compare your belief-state diagram with both the causal-state automaton constructed in Part 2 and the three-state generative HMM from Exercise 1.

(i) Match each belief to the causal state with the same shortest-length representative. Do the symbol-labelled transitions agree?

(ii) Which beliefs correspond to knowing the hidden phase exactly, and which represent uncertainty over several phases? Show that the hidden phase becomes known after at most three observations and remains known thereafter.

(iii) Observe that the finite-history belief-state presentation has seven states while the generative HMM has only three. What does this suggest about the possible relative difficulty of generation and inference?

(d) For each belief $\eta^{(x_{1:n})} \in \mathcal{B}$ from part (b), compute its associated next-token probability vector

$$p^{(x_{1:n})} := (\Pr(X_{n+1} = a \mid X_{1:n} = x_{1:n}))_{a\in\{0,1\}} = \eta^{(x_{1:n})}A, \qquad A_{ia} := \sum_j T_{ij}^{(a)}.$$

Visualise the belief geometry on the 2-simplex over hidden states $(0, 1, R)$ and the next-token geometry on the 1-simplex over symbols $(0, 1)$.

(e) Calculate

$$\Pr(X_{3:4} = 00 \mid X_{1:2} = 10) \qquad \text{and} \qquad \Pr(X_{3:4} = 00 \mid X_{1:2} = 01).$$

Deduce that the next-token probability vector need not be sufficient for predicting the entire future.

(f) **Extension.** Let $\delta := \eta^{(10)} - \eta^{(01)}$. Show that

$$\delta A = 0 \qquad \text{but} \qquad \delta T^{(0)}T^{(0)}\mathbf{1} \neq 0.$$

Interpret these two statements in terms of one-step and longer-horizon prediction.

Solution.

(a) Multiplying the initial belief by the two symbol matrices gives

$$\eta^{(\emptyset)}T^{(0)} = \begin{bmatrix}(1-p)/3 & 1/3 & 0\end{bmatrix}, \qquad \eta^{(\emptyset)}T^{(1)} = \begin{bmatrix}p/3 & 0 & 1/3\end{bmatrix}.$$

Their row sums are respectively $(2-p)/3$ and $(1+p)/3$. Normalising therefore gives

$$\eta^{(0)} = \begin{bmatrix}\frac{1-p}{2-p} & \frac{1}{2-p} & 0\end{bmatrix}, \qquad \eta^{(1)} = \begin{bmatrix}\frac{p}{1+p} & 0 & \frac{1}{1+p}\end{bmatrix}.$$

After observing 0, the current phase is either 0 or 1 with the displayed posterior probabilities, and cannot be R. After observing 1, it is either 0 or R, and cannot be 1.

(b) Direct evaluation gives

$$\eta^{(00)} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}, \qquad \eta^{(01)} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix},$$
$$\eta^{(10)} = \begin{bmatrix} 1-p & p & 0 \end{bmatrix}, \quad \eta^{(11)} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}.$$

Thus the pure beliefs are $\eta^{(11)}$, $\eta^{(00)}$, and $\eta^{(01)}$. The pure beliefs are distinct. The beliefs $\eta^{(\emptyset)}$, $\eta^{(0)}$, and $\eta^{(1)}$ have different supports from one another and from the pure beliefs. The only remaining comparison is between $\eta^{(0)}$ and $\eta^{(10)}$, which both have support $\{0, 1\}$. Equality would require $1/(2-p) = p$, or $(1-p)^2 = 0$, contrary to $p < 1$.

The allowable updates are

current belief	0	1
$\eta^{(\emptyset)}$	$\eta^{(0)}$	$\eta^{(1)}$
$\eta^{(0)}$	$\eta^{(00)}$	$\eta^{(01)}$
$\eta^{(1)}$	$\eta^{(10)}$	$\eta^{(11)}$
$\eta^{(10)}$	$\eta^{(00)}$	$\eta^{(01)}$
$\eta^{(11)}$	$\eta^{(00)}$	impossible
$\eta^{(00)}$	impossible	$\eta^{(01)}$
$\eta^{(01)}$	$\eta^{(11)}$	$\eta^{(11)}$

This table specifies the requested symbol-labelled diagram and shows that every allowable update stays in $\mathcal{B}$.

(c) (i) Under the assumptions used in Part 2, the correspondence is

$$\epsilon(\emptyset) \longleftrightarrow \eta^{(\emptyset)}, \qquad \epsilon(0) \longleftrightarrow \eta^{(0)}, \qquad \epsilon(1) \longleftrightarrow \eta^{(1)},$$
$$\epsilon(10) \longleftrightarrow \eta^{(10)}, \quad \epsilon(11) \longleftrightarrow \eta^{(11)}, \quad \epsilon(00) \longleftrightarrow \eta^{(00)}, \quad \epsilon(01) \longleftrightarrow \eta^{(01)}.$$

The transition table from part (b) agrees with the causal-state automaton after making these substitutions.

(ii) The pure beliefs $\eta^{(11)}$, $\eta^{(00)}$, and $\eta^{(01)}$ identify the hidden phase exactly. The other four beliefs assign positive probability to more than one phase and therefore represent uncertainty. After two observations only the history 10 leaves a mixed belief. Either allowable third symbol sends $\eta^{(10)}$ to a pure belief. Once the belief is pure, every allowable transition in the table leads to another pure belief, so the phase remains known.

(iii) The generator has access to its actual hidden phase and needs only the three states 0, 1, and R. An observer begins with an unknown phase and must also represent four transient posterior distributions before synchronising with the generator. This example therefore shows that inference can require a richer state description than a particular generative presentation.

(d) The probability of each symbol from a hidden phase is obtained by summing the corresponding row of its symbol matrix over destination states. With column order $(0, 1)$, this gives

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1-p & p \end{bmatrix}.$$

Hence

belief	next-token vector
$\eta^{(\emptyset)}$	$p^{(\emptyset)} = \left(\dfrac{2-p}{3}, \dfrac{1+p}{3}\right)$
$\eta^{(0)}$	$p^{(0)} = \left(\dfrac{1-p}{2-p}, \dfrac{1}{2-p}\right)$
$\eta^{(1)}$	$p^{(1)} = \left(\dfrac{1}{1+p}, \dfrac{p}{1+p}\right)$
$\eta^{(00)}$	$p^{(00)} = (0, 1)$
$\eta^{(01)}$	$p^{(01)} = (1-p, p)$
$\eta^{(10)}$	$p^{(10)} = (1-p, p)$
$\eta^{(11)}$	$p^{(11)} = (1, 0)$

On the belief 2-simplex, $\eta^{(11)}$, $\eta^{(00)}$, and $\eta^{(01)}$ are the three vertices, while $\eta^{(\emptyset)}$ is the barycentre. The beliefs $\eta^{(0)}$ and $\eta^{(10)}$ lie on the edge joining $\eta^{(11)}$ to $\eta^{(00)}$, and $\eta^{(1)}$ lies on the edge joining $\eta^{(11)}$ to $\eta^{(01)}$.

On the next-token 1-simplex, each point is located by the second coordinate in the table, namely its probability of emitting 1. The relation $p^{(x_{1:n})} = \eta^{(x_{1:n})}A$ maps the two distinct beliefs $\eta^{(01)}$ and $\eta^{(10)}$ to the same point $p^{(01)} = p^{(10)} = (1-p, p)$.

(e) Starting from $\eta^{(10)} = (1-p, p, 0)$, observing a 0 can only come from phase 0 and moves the process to phase 1, which cannot emit a second 0. Hence

$$\Pr(X_{3:4} = 00 \mid X_{1:2} = 10) = \eta^{(10)}T^{(0)}T^{(0)}\mathbf{1} = 0.$$

Starting from $\eta^{(01)} = (0, 0, 1)$, the first 0 has probability $1-p$ and moves the process to phase 0, which emits the second 0 deterministically. Therefore

$$\Pr(X_{3:4} = 00 \mid X_{1:2} = 01) = \eta^{(01)}T^{(0)}T^{(0)}\mathbf{1} = 1-p.$$

The histories agree about the next symbol but disagree about a two-symbol continuation, so their shared next-token vector is not sufficient for the entire future.

(f) We have

$$\delta = \begin{bmatrix} 1-p & p & -1 \end{bmatrix}.$$

Direct multiplication gives

$$\delta A = \begin{bmatrix} 0 & 0 \end{bmatrix}, \qquad \delta T^{(0)}T^{(0)}\mathbf{1} = -(1-p) \neq 0.$$

Thus the difference between the beliefs is invisible to the one-step readout but visible to a readout associated with the two-symbol future 00.

Exercise 3

Nonunifilar generators and large belief-state automata

For a belief η and a symbol a satisfying $\eta T^{(a)}\mathbf{1} > 0$, write

$$F_a(\eta) := \frac{\eta T^{(a)}}{\eta T^{(a)}\mathbf{1}}.$$

Thus the belief-state automaton contains the transition

$$\eta \xrightarrow{a:\eta T^{(a)}\mathbf{1}} F_a(\eta).$$

(a) **The Simple Nonunifilar Source.** Revisit the source from Part 2. With state order (A, B), its symbol matrices and initial belief are

$$T^{(0)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}, \qquad T^{(1)} = \begin{bmatrix} 0 & 0 \\ \frac{1}{2} & 0 \end{bmatrix}, \qquad \eta^{(\emptyset)} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

(i) For $n \geq 0$, show that

$$\eta^{(10^n)} = \begin{bmatrix} \dfrac{1}{n+1} & \dfrac{n}{n+1} \end{bmatrix}, \qquad \eta^{(0^n)} = \begin{bmatrix} \dfrac{1}{n+2} & \dfrac{n+1}{n+2} \end{bmatrix}.$$

Deduce that $\eta^{(0^n)} = \eta^{(10^{n+1})}$ and observe that $\eta^{(\emptyset)} = \eta^{(10)}$.

(ii) Show that, for every $n \geq 1$,
$$\eta^{(10^n 1)} = \eta^{(1)}.$$

(iii) Draw the first four states of the belief-state automaton and indicate its continuing structure with an ellipsis. Show that the beliefs $\eta^{(10^n)}$ are all distinct, identify their limiting point in the 1-simplex, and explain why the two-state generator produces a countably infinite belief-state automaton.

(b) **Mess3.** Now take hidden-state and symbol sets both equal to $\{0, 1, 2\}$, with

$$\alpha = \frac{1}{5}, \qquad b = \frac{1-\alpha}{2} = \frac{2}{5}, \qquad x = \frac{3}{20}, \qquad y = 1 - 2x = \frac{7}{10},$$

and

$$T^{(0)} = \begin{bmatrix} \alpha y & bx & bx \\ \alpha x & by & bx \\ \alpha x & bx & by \end{bmatrix}, \quad T^{(1)} = \begin{bmatrix} by & \alpha x & bx \\ bx & \alpha y & bx \\ bx & \alpha x & by \end{bmatrix},$$

$$T^{(2)} = \begin{bmatrix} by & bx & \alpha x \\ bx & by & \alpha x \\ bx & bx & \alpha y \end{bmatrix}, \quad \eta^{(\emptyset)} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}.$$

(i) Verify that $T = \sum_{a=0}^{2} T^{(a)}$ is row-stochastic and explain why this generator is nonunifilar. Compute $\eta^{(0)}, \eta^{(1)}$, and $\eta^{(2)}$.

(ii) Compute $\eta^{(00)}, \eta^{(01)}$, and $\eta^{(02)}$. Use the cyclic symmetry of the process to sketch the first two generations of the belief-state automaton, including the symbol labels.

(iii) Compute the next-token matrix A, with $A_{ia} = \sum_j T^{(a)}_{ij}$, and show that it is invertible. What does this imply about whether distinct beliefs can be merged by the map $p^{(x_{1:n})} = \eta^{(x_{1:n})} A$? Contrast this with Exercise 2.

(iv) Using a short program, form

$$\mathcal{B}_d := \{\eta^{(x_{1:d})} : x_{1:d} \in \{0, 1, 2\}^d\}, \qquad 0 \leq d \leq 7.$$

Record the number of distinct beliefs at each depth. Visualise $\mathcal{B}_7$ on the hidden-state 2-simplex, colouring each point by its final symbol. Also visualise the associated next-token vectors on the symbol 2-simplex. Describe the geometric structure that emerges and how the three maps F_0, F_1, F_2 generate the outgoing transitions of the belief-state automaton.

(v) Explain why the belief-state automata constructed in this exercise are unifilar even though their generators are not. Compare the ladder-with-resets structure of the Simple Nonunifilar Source with the branching structure of Mess3. Finally, explain why the set of beliefs reached by finite histories is countable even though its closure can have a much richer fractal geometry.

Solution.

(a) (i) After observing 1, the hidden state is A. A path emitting the following 0^n either remains in A throughout or moves from A to B after one of the n emissions. These $n+1$ paths have equal weight 2^{-n}; one ends in A and n end in B. Thus

$$\eta^{(1)}(T^{(0)})^n = 2^{-n}\begin{bmatrix}1 & n\end{bmatrix}, \qquad \eta^{(10^n)} = \begin{bmatrix}\frac{1}{n+1} & \frac{n}{n+1}\end{bmatrix}.$$

Starting instead from the uniform initial belief gives

$$\eta^{(\emptyset)}(T^{(0)})^n = 2^{-(n+1)}\begin{bmatrix}1 & n+1\end{bmatrix}, \qquad \eta^{(0^n)} = \begin{bmatrix}\frac{1}{n+2} & \frac{n+1}{n+2}\end{bmatrix}.$$

Hence $\eta^{(0^n)} = \eta^{(10^{n+1})}$, and the case $n=0$ gives $\eta^{(\emptyset)} = \eta^{(10)}$.

(ii) For $n \geq 1$, direct multiplication gives

$$\eta^{(10^n)}T^{(1)} = \begin{bmatrix}\frac{n}{2(n+1)} & 0\end{bmatrix}.$$

Normalising yields $\eta^{(10^n1)} = \begin{bmatrix}1 & 0\end{bmatrix} = \eta^{(1)}$.

(iii) The automaton begins

$$\eta^{(1)} \xrightarrow{0} \eta^{(10)} \xrightarrow{0} \eta^{(100)} \xrightarrow{0} \eta^{(1000)} \xrightarrow{0} \cdots,$$

and every $\eta^{(10^n)}$ with $n \geq 1$ has a 1-transition back to $\eta^{(1)}$. Since the first coordinate $1/(n+1)$ is different for each n, the beliefs are distinct, and

$$\lim_{n\to\infty} \eta^{(10^n)} = \begin{bmatrix}0 & 1\end{bmatrix}.$$

This limiting belief is not reached after any finite history. Hence the two hidden generator states induce countably infinitely many reachable beliefs.

(b) (i) Substitution gives $\alpha y = 7/50$, $bx = 3/50$, $\alpha x = 3/100$, and $by = 7/25$. Each entry is positive, and every state has several possible destinations for each emitted symbol, so the generator is nonunifilar. The rows of $T^{(0)} + T^{(1)} + T^{(2)}$ each sum to one. By multiplying the uniform belief and normalising,

$$\eta^{(0)} = \begin{bmatrix}\frac{1}{5} & \frac{2}{5} & \frac{2}{5}\end{bmatrix}, \quad \eta^{(1)} = \begin{bmatrix}\frac{2}{5} & \frac{1}{5} & \frac{2}{5}\end{bmatrix}, \quad \eta^{(2)} = \begin{bmatrix}\frac{2}{5} & \frac{2}{5} & \frac{1}{5}\end{bmatrix}.$$

(ii) Updating $\eta^{(0)}$ by each symbol gives

$$\eta^{(00)} = \begin{bmatrix}\frac{13}{87} & \frac{37}{87} & \frac{37}{87}\end{bmatrix}, \quad \eta^{(01)} = \begin{bmatrix}\frac{52}{163} & \frac{37}{163} & \frac{74}{163}\end{bmatrix}, \quad \eta^{(02)} = \begin{bmatrix}\frac{52}{163} & \frac{74}{163} & \frac{37}{163}\end{bmatrix}.$$

Cyclically permuting both the state coordinates and symbol labels gives the remaining six depth-two beliefs. The root has three symbol-labelled children, and each of those children again has three symbol-labelled successors.

(iii) Summing over destination states gives

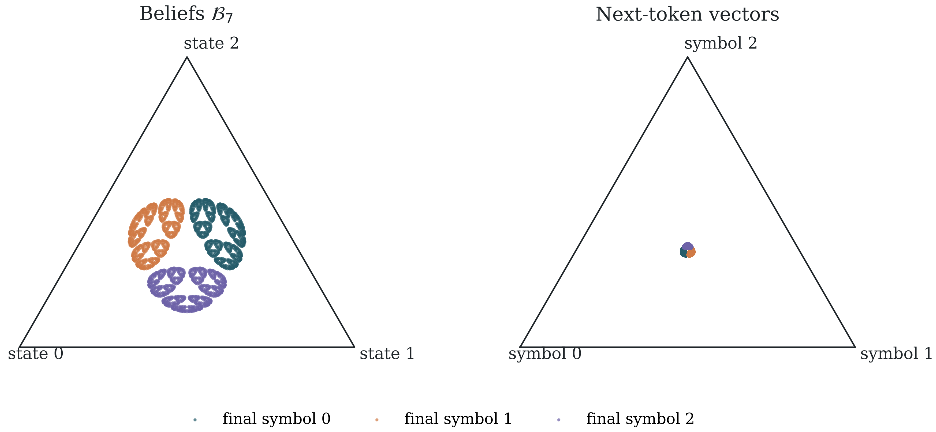
$$A = \begin{bmatrix}\frac{13}{50} & \frac{37}{100} & \frac{37}{100} \\ \frac{37}{100} & \frac{13}{50} & \frac{37}{100} \\ \frac{37}{100} & \frac{37}{100} & \frac{13}{50}\end{bmatrix}.$$

Its eigenvalues are $1, -11/100, -11/100$, so $\det A = 121/10000 \neq 0$. Consequently $\eta A = \eta' A$ implies $\eta = \eta'$: the next-token readout does not merge distinct Mess3 beliefs. In Exercise 2 the corresponding matrix has a nontrivial kernel and maps $\eta^{(01)}$ and $\eta^{(10)}$ to the same next-token vector.

(iv) For these parameters, numerical enumeration gives

d	0	1	2	3	4	5	6	7
$\lvert\mathcal{B}_d\rvert$	1	3	9	27	81	243	729	2187

with no coincidences at the displayed depths. The plots show three recursively repeated clusters forming a fractal pattern. The next-token plot is an invertible linear image of the belief plot, so it preserves the distinct points and their recursive organisation.



At a point η, the three outgoing edges are obtained by applying F_0, F_1, F_2 and have probabilities given by the three coordinates of ηA. Thus the three geometric copies visible at the next depth are also the three symbol-labelled branches of the belief-state automaton.

(v) A belief and the observed symbol determine a unique updated belief $F_a(\eta)$, so the belief-state presentation is unifilar. The original generator is nonunifilar because its current hidden state and emitted symbol can leave several possible next hidden states. For the Simple Nonunifilar Source the reachable beliefs form a one-dimensional ladder, with each 1-edge resetting to η_0. For Mess3 every symbol remains possible and the three update maps generate a rapidly branching, self-similar geometry. Finally, finite words form a countable set, so their image under the belief map is countable. Taking the closure adds limiting beliefs and can produce an uncountable fractal set.

Exercise 4

Generalised hidden Markov models and predictive vectors

A d-dimensional generalised hidden Markov model (GHMM) consists of a finite alphabet $\mathcal{X}$, real $d \times d$ matrices $(T^{(a)})_{a\in\mathcal{X}}$, an initial row vector $\eta^{(\emptyset)}$, and a column vector ϕ. Writing

$$T := \sum_{a\in\mathcal{X}} T^{(a)},$$

these objects satisfy

$$T\phi = \phi, \qquad \eta^{(\emptyset)}\phi = 1, \qquad \eta^{(\emptyset)}T^{(x_1)}\cdots T^{(x_n)}\phi \geq 0$$

for every $x_{1:n} \in \mathcal{X}^n$ and every $n \geq 0$.

(a) **A probability law from matrix products.** Consider

$$\Pr(X_{1:n} = x_{1:n}) = \eta^{(\emptyset)}T^{(x_1)}\cdots T^{(x_n)}\phi.$$

(i) Show that these quantities are nonnegative and that

$$\sum_{x_{1:n}\in\mathcal{X}^n} \Pr(X_{1:n} = x_{1:n}) = 1$$

for every $n \geq 0$.

(ii) Show that

$$\sum_{a\in\mathcal{X}} \Pr(X_{1:n+1} = x_{1:n}a) = \Pr(X_{1:n} = x_{1:n}).$$

Explain why this is the consistency condition needed to define a one-sided stochastic process.

(iii) The GHMM conditions do not require $\eta^{(\emptyset)}T = \eta^{(\emptyset)}$. Explain why the process need not therefore be stationary. Show that this additional equality is sufficient for stationarity.

(b) **Predictive vectors.** For any $x_{1:n}$ satisfying $\Pr(X_{1:n} = x_{1:n}) > 0$, define its predictive vector by

$$\eta^{(x_{1:n})} := \frac{\eta^{(\emptyset)}T^{(x_1)}\cdots T^{(x_n)}}{\eta^{(\emptyset)}T^{(x_1)}\cdots T^{(x_n)}\phi}.$$

(i) Show that

$$\eta^{(x_{1:n})}\phi = 1$$

and, for every continuation $y_{1:m}$,

$$\Pr(X_{n+1:n+m} = y_{1:m} \mid X_{1:n} = x_{1:n}) = \eta^{(x_{1:n})}T^{(y_1)}\cdots T^{(y_m)}\phi.$$

(ii) Derive the recursive update and identify its normalising factor:

$$\eta^{(x_{1:n}a)} = \frac{\eta^{(x_{1:n})}T^{(a)}}{\eta^{(x_{1:n})}T^{(a)}\phi}, \qquad \Pr(X_{n+1} = a \mid X_{1:n} = x_{1:n}) = \eta^{(x_{1:n})}T^{(a)}\phi.$$

(iii) Conceptually contrast an HMM belief state with a GHMM predictive vector. Compare the meaning of their coordinates, their positivity and normalisation properties, and how they encode predictions of future observations.

(c) **Predictive vectors need not be probability vectors.** Consider the two-state HMM

$$M^{(0)} = \begin{bmatrix} \frac{4}{5} & 0 \\ 0 & \frac{1}{5} \end{bmatrix}, \qquad M^{(1)} = \begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{4}{5} \end{bmatrix}, \qquad \alpha = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Thus a hidden state is selected initially and thereafter emits a biased coin without changing state. Let

$$S = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}, \qquad T^{(a)} := S^{-1}M^{(a)}S, \qquad \eta^{(\emptyset)} := \alpha S, \qquad \phi := S^{-1}\mathbf{1}.$$

(i) Compute $T^{(0)}, T^{(1)}, \eta^{(\emptyset)}$, and ϕ. Show that, for every $x_{1:n}$,

$$\eta^{(\emptyset)}T^{(x_1)}\cdots T^{(x_n)}\phi = \alpha M^{(x_1)}\cdots M^{(x_n)}\mathbf{1}.$$

Deduce that this change of coordinates gives a valid GHMM for the same process.

(ii) Compute $\eta^{(0)}$ and $\eta^{(1)}$. Verify directly from $\eta^{(0)}$ that

$$\Pr(X_2 = 0 \mid X_1 = 0) = \frac{17}{25}, \qquad \Pr(X_2 = 1 \mid X_1 = 0) = \frac{8}{25}.$$

Explain why the entries of a GHMM predictive vector need not themselves be probabilities.

(d) **The probability matrix and minimal dimension.** For finite sets $\mathcal{U}, \mathcal{V} \subseteq \mathcal{X}^*$, define the probability matrix by

$$\mathsf{P}_{\mathcal{U},\mathcal{V}}(u,v) := \Pr(X_{1:|u|+|v|} = uv), \qquad u \in \mathcal{U}, \quad v \in \mathcal{V}.$$

(i) Use the GHMM matrix product to factor $\mathsf{P}_{\mathcal{U},\mathcal{V}}$ as a matrix of prefix row vectors times a matrix of continuation column vectors. Deduce that

$$\operatorname{rank} \mathsf{P}_{\mathcal{U},\mathcal{V}} \le d$$

for every d-dimensional GHMM realisation.

(ii) For the process in part (c), take $\mathcal{U} = \mathcal{V} = \{\emptyset, 0, 1\}$. Compute $\mathsf{P}_{\mathcal{U},\mathcal{V}}$ and use an SVD to find its singular values. Deduce that the two-dimensional GHMM is minimal-dimensional.

Solution.

(a) (i) Nonnegativity is one of the GHMM assumptions. Expanding the product of the sum of the symbol matrices gives

$$\sum_{x_{1:n} \in \mathcal{X}^n} T^{(x_1)} \cdots T^{(x_n)} = T^n.$$

Hence

$$\sum_{x_{1:n} \in \mathcal{X}^n} \Pr(X_{1:n} = x_{1:n}) = \eta^{(\emptyset)} T^n \phi = \eta^{(\emptyset)} \phi = 1,$$

where $T^n \phi = \phi$ follows from $T\phi = \phi$. For $n = 0$ this also says that the empty history has probability one.

(ii) Summing over the final symbol gives

$$\begin{aligned}
\sum_{a \in \mathcal{X}} \Pr(X_{1:n+1} = x_{1:n}a) &= \eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)} \left(\sum_{a \in \mathcal{X}} T^{(a)} \right) \phi \\
&= \eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)} T \phi \\
&= \Pr(X_{1:n} = x_{1:n}).
\end{aligned}$$

Thus the length-n distribution is the marginal of the length-$(n+1)$ distribution over its final coordinate, as required for a process indexed by $1, 2, \ldots$.

(iii) The displayed GHMM conditions relate distributions at successive lengths but do not say that a block has the same distribution after shifting its time indices. If $\eta^{(\emptyset)} T = \eta^{(\emptyset)}$, then

$$\begin{aligned}
\sum_{a \in \mathcal{X}} \Pr(X_{1:n+1} = a x_{1:n}) &= \eta^{(\emptyset)} T T^{(x_1)} \cdots T^{(x_n)} \phi \\
&= \eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)} \phi \\
&= \Pr(X_{1:n} = x_{1:n}).
\end{aligned}$$

The left side is $\Pr(X_{2:n+1} = x_{1:n})$, so the block distributions are invariant under a one-step shift. Iteration gives stationarity.

(b) (i) The denominator in the definition is $\Pr(X_{1:n} = x_{1:n})$, and so

$$\eta^{(x_{1:n})} \phi = 1.$$

By the definition of conditional probability,

$$\begin{aligned}
&\Pr(X_{n+1:n+m} = y_{1:m} \mid X_{1:n} = x_{1:n}) \\
&= \frac{\eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)} T^{(y_1)} \cdots T^{(y_m)} \phi}{\eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)} \phi} \\
&= \eta^{(x_{1:n})} T^{(y_1)} \cdots T^{(y_m)} \phi.
\end{aligned}$$

(ii) Taking $m = 1$ in part (i) gives

$$\Pr(X_{n+1} = a \mid X_{1:n} = x_{1:n}) = \eta^{(x_{1:n})} T^{(a)} \phi, \qquad \eta^{(x_{1:n}a)} = \frac{\eta^{(x_{1:n})} T^{(a)}}{\eta^{(x_{1:n})} T^{(a)} \phi}.$$

The first quantity is exactly the normalising factor in the update shown by the second equality.

(iii) An HMM belief state has coordinates $\Pr(S_n = i \mid X_{1:n} = x_{1:n})$, so they are nonnegative, sum to one, and refer to mutually exclusive hidden states. Predictive vectors are coordinates in a linear realisation: they need only satisfy $\eta^{(x_{1:n})}\phi = 1$ and may have negative entries. Both summarise the history sufficiently to compute future-word probabilities through symbol-matrix products.

(c) (i) Since

$$S^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ 0 & 1 \end{bmatrix},$$

direct calculation gives

$$T^{(0)} = \begin{bmatrix} \frac{4}{5} & -\frac{3}{10} \\ 0 & \frac{1}{5} \end{bmatrix}, \qquad T^{(1)} = \begin{bmatrix} \frac{1}{5} & \frac{3}{10} \\ 0 & \frac{4}{5} \end{bmatrix},$$

and

$$\eta^{(\emptyset)} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad \phi = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

In particular $T = T^{(0)} + T^{(1)} = I$, so $T\phi = \phi$, and $\eta^{(\emptyset)}\phi = 1$. In a product, adjacent factors SS^{-1} cancel:

$$\begin{aligned} \eta^{(\emptyset)} T^{(x_1)} \cdots T^{(x_n)} \phi &= \alpha S (S^{-1} M^{(x_1)} S) \cdots (S^{-1} M^{(x_n)} S) S^{-1} \mathbf{1} \\ &= \alpha M^{(x_1)} \cdots M^{(x_n)} \mathbf{1}. \end{aligned}$$

The right side is an HMM sequence probability and is therefore nonnegative. All GHMM conditions are satisfied, and both realisations define the same process.

(ii) Both first-symbol probabilities are 1/2. Normalising gives

$$\eta^{(0)} = \begin{bmatrix} \frac{8}{5} & -\frac{3}{5} \end{bmatrix}, \qquad \eta^{(1)} = \begin{bmatrix} \frac{2}{5} & \frac{3}{5} \end{bmatrix}.$$

Moreover,

$$T^{(0)}\phi = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{5} \end{bmatrix}, \qquad T^{(1)}\phi = \begin{bmatrix} \frac{1}{2} \\ \frac{4}{5} \end{bmatrix}.$$

Hence

$$\eta^{(0)} T^{(0)} \phi = \frac{17}{25}, \qquad \eta^{(0)} T^{(1)} \phi = \frac{8}{25}.$$

Although the second coordinate of $\eta^{(0)}$ is negative, all continuation probabilities obtained from it are valid. The predictive vector consists of coordinates in a chosen linear representation; unlike an HMM belief, its entries do not have to describe mutually exclusive hidden events.

(d) (i) For $u = u_1 \cdots u_r$ and $v = v_1 \cdots v_s$,

$$\mathsf{P}_{\mathcal{U},\mathcal{V}}(u, v) = \big(\eta^{(\emptyset)} T^{(u_1)} \cdots T^{(u_r)}\big)\big(T^{(v_1)} \cdots T^{(v_s)} \phi\big).$$

Collecting the first factors as rows and the second factors as columns gives the requested factorisation through $\mathbb{R}^d$. The rank of the product is therefore at most d.

(ii) The required probabilities are

$$\begin{gathered} \Pr(X_1 = 0) = \Pr(X_1 = 1) = \frac{1}{2}, \\ \Pr(X_{1:2} = 00) = \Pr(X_{1:2} = 11) = \frac{17}{50}, \quad \Pr(X_{1:2} = 01) = \Pr(X_{1:2} = 10) = \frac{4}{25}. \end{gathered}$$

Thus, in the order $(\emptyset, 0, 1)$,

$$\mathsf{P}_{\mathcal{U},\mathcal{V}} = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{17}{50} & \frac{4}{25} \\ \frac{1}{2} & \frac{4}{25} & \frac{17}{50} \end{bmatrix}, \qquad \sigma(\mathsf{P}_{\mathcal{U},\mathcal{V}}) = \left(\frac{3}{2}, \frac{9}{50}, 0\right).$$

It therefore has rank two. Every GHMM realisation must have dimension at least two, while part (c) supplies a two-dimensional realisation. It is minimal-dimensional.